SYSTEMS OF FIRST-ORDER LOGIC

1. INTRODUCTION

In these notes we present first-order logic from syntactic and proof-theoretic perspective. After discussing syntax of first-order languages we introduce a Hilbert-style deductive system for minimal logic and its extensions by propositional axiom schemata. A key property of the system is that it has modus ponens as the unique inference rule. All other logical behavior is encoded via axiom schemata. In fourth section we establish basic meta-theorems. Next sections are devoted to conservative extensions of first-order theories. We discuss three such extensions: relation symbols, indefinite descriptions of individuals, definite descriptions for functions. In the last section we introduce intuitionistic and classical logic.

Throughout these notes we fix five elements

$$\bot, \rightarrow, \wedge, \vee, \forall, \exists$$

and we call them *logical constants*. Meta-theoretic equality is denoted by $\equiv$.

2. FIRST-ORDER LANGUAGES AND THEIR SYNTAX

Definition 2.1. A *signature* consists of the following data.

- A collection of *n-ary relation symbols* for every positive natural number n.
- A collection of *n-ary function symbols* for every natural number n.
- Optionally it may contain the *equality symbol* $=$.

Moreover, nullary function symbols are called *constants*.

Definition 2.2. A *first-order language* consists of a signature (potentially with equality) and an infinite collection of variables.

In what follows we fix a first-order language $\mathcal{L}$.

Definition 2.3. Words over the alphabet consisting of parentheses, logical constants, the signature of $\mathcal{L}$ and variables of $\mathcal{L}$ are *expressions* of $\mathcal{L}$.

Definition 2.4. The collection *terms* of $\mathcal{L}$ is the smallest collection of expressions of $\mathcal{L}$ such that the following conditions holds.

(1) Every variable of $\mathcal{L}$ is a term.

(2) If $t_1, ..., t_n$ are terms of $\mathcal{L}$ and f is n-ary function symbol, then $ft_1...t_n$ is a term of $\mathcal{L}$

Definition 2.5. An *atomic formula* of $\mathcal{L}$ is an expression of $\mathcal{L}$ of the following two forms.

(1) $t_1 = t_2$ for terms t_1, t_2 of $\mathcal{L}$ if the equality is included of $\mathcal{L}$.

(2) $rt_1...t_n$ where r is an n-ary relation symbol and $t_1, ..., t_n$ are terms of $\mathcal{L}$.

Definition 2.6. The class of *formulas* of $\mathcal{L}$ is the smallest class of expressions of $\mathcal{L}$ such that the following conditions hold.

(1) Every atomic formula is a formula of $\mathcal{L}$.

(2) $\bot$ is a formula of $\mathcal{L}$.

(3) If ϕ and ψ are formulas of $\mathcal{L}$, then

$$(\phi \to \psi),\ (\phi \wedge \psi),\ (\phi \vee \psi)$$

are formulas of $\mathcal{L}$.

(4) If x is a variable and ϕ is a formula of $\mathcal{L}$, then

$$\forall_x \phi,\ \exists_x \phi$$

are formulas of $\mathcal{L}$.

Remark 2.7. Let ϕ, ψ be formulas of $\mathcal{L}$. Then we define

$$(\neg\phi) \equiv (\phi \to \bot),\ (\phi \leftrightarrow \psi) \equiv ((\phi \to \psi) \wedge (\psi \to \phi))$$

The following results study the determinacy of syntax of $\mathcal{L}$.

Theorem 2.8. *The following assertions hold.*

(1) *Let t be a term of $\mathcal{L}$ and let s be its proper segment. If s is a term of $\mathcal{L}$, then s coincides with t.*

(2) *Let γ be a formula of $\mathcal{L}$ and let θ be its initial proper segment. If θ is a formula of $\mathcal{L}$, then θ coincides with γ.*

For the proof we need some additional result.

Lemma 2.8.1. *Let θ be a formula of $\mathcal{L}$. Then numbers of occurrences of left parentheses and right parentheses in θ are the same.*

Proof of the lemma. Follows by straightforward induction on the complexity θ. □

Proof of the theorem. The proof of **(1)** goes by the induction on the complexity of t.

- If t is a variable, then the assertion holds.
- Suppose that t is of the form $ft_1...t_n$ where f is n-ary function symbol of $\mathcal{L}$ and $t_1, ..., t_n$ are terms of $\mathcal{L}$. Since s is a term, it is nonempty. Hence s starts with f. It follows that s is of the form $fs_1...s_n$ for some terms $s_1, ..., s_n$ of $\mathcal{L}$. Suppose that s_i coincides with t_i for $i < j$ for some $j < n$. Then either s_j is a proper segment of t_j or t_j is a proper segment of s_j. In either case the induction assumption implies that s_j coincides with t_j. This proves that s_i coincides with t_i for all $i \leq n$. Therefore, s and t coincide.

The proof of **(2)** goes by the induction on the complexity of γ.

- If γ is $\bot$, then the assertion holds.
- Suppose that γ is of the form $rt_1...t_n$ where r is n-ary relation symbol of $\mathcal{L}$ and $t_1, ..., t_n$ are terms of $\mathcal{L}$. Since θ is a formula of $\mathcal{L}$, it is nonempty. Hence θ starts with r. It follows that θ is of the form $rs_1...s_n$ for some terms $s_1, ..., s_n$ of $\mathcal{L}$. Suppose that s_i coincides with t_i for $i < j \leq n$. Then either s_j is a proper segment of t_j or t_j is a proper segment of s_j. In either case by Lemma **(1)** it follows that s_j coincides with t_j. This proves that s_i coincides with t_i for all $i \leq n$. Therefore, θ and γ coincide.
- Suppose that γ is of the form $t_1 = t_2$ for some terms t_1, t_2 of $\mathcal{L}$. Then θ is of the form $t_1 = s$ for some term s of $\mathcal{L}$. Since s is an initial segment of t_2, we derive by **(1)** that s and t_2 coincide and hence θ coincides with γ.
- Suppose that γ has one of forms

$$(\phi \to \psi),\ (\phi \wedge \psi),\ (\phi \vee \psi)$$

for some formulas ϕ, ψ of $\mathcal{L}$. Then according to Lemma 2.8.1 we infer that θ coincides with γ.

- Suppose that γ is of the form $\forall_x \phi$ for some variable x and formula ϕ of $\mathcal{L}$. Then θ is of the form $\forall_x \psi$ for a formula ψ of $\mathcal{L}$. Thus ψ is an initial segment of ϕ. Hence by induction hypothesis ψ coincides with ϕ. Therefore, θ coincides with γ. Analogical argument works if γ is of the form $\exists_x \phi$ for some variable x and formula ϕ of $\mathcal{L}$.

□

Theorem 2.9 [Unique readability of terms and formulas]. *The following assertions hold.*

(1) *Let t be a term of $\mathcal{L}$. Then either t is a variable or there exists a unique n-ary function symbol f and unique terms $t_1, ..., t_n$ such that*

$$t = f t_1 ... t_n$$

for some $n \in \mathbb{N}$.

(2) *Let γ be an atomic formula of $\mathcal{L}$. Then either there are unique terms t_1, t_2 of $\mathcal{L}$ such that γ is $t_1 = t_2$ or there exists a unique n-ary relation symbol r and unique terms $t_1, ..., t_n$ such that γ is $r t_1 ... t_n$ for some $n \in \mathbb{N}$.*

(3) *Let γ be a formula of $\mathcal{L}$. Then precisely one of the assertions hold.*

- *γ is atomic.*
- *γ is $\bot$.*
- *γ is $(\phi \to \psi)$ for unique formulas ϕ, ψ of $\mathcal{L}$.*
- *γ is $(\phi \wedge \psi)$ for unique formulas ϕ, ψ of $\mathcal{L}$.*
- *γ is $(\phi \vee \psi)$ for unique formulas ϕ, ψ of $\mathcal{L}$.*
- *γ is $\forall_x \phi$ for a unique variable x and a unique formula ϕ of $\mathcal{L}$.*
- *γ is $\exists_x \phi$ for a unique variable x and a unique formula ϕ of $\mathcal{L}$.*

Proof. Note that **(1)** is straightforward if term is a variable. Suppose that

$$f t_1 ... t_n \equiv g s_1 ... s_m$$

where f is an n-ary function symbol, g is m-ary function symbol and $t_1, ..., t_n, s_1, ..., s_m$ are terms. Then f coincides with g and thus n equals m. Next assume that t_i coincides with s_i for $i < j \leq n$. Then either t_j is an initial segment of s_j or s_j is an initial segment of t_j. In either case Theorem 2.8 implies that t_j coincides with s_j. This proves that t_i coincides with s_i for all $i = 1, ..., n$. This completes the proof of **(1)**.

For the proof of **(2)** note that each atomic formula either starts with relational symbol or from function symbol. Moreover, this two cases are exclusive. In the first case the proof is analogical to the one of **(1)**. Suppose that an atomic formula starts with function symbol. Then it must contain the equality sign. Suppose that

$$t_1 = t_2,\ s_1 = s_2$$

are two representations of the same atomic formula for terms t_1, t_2, s_1, s_2 of $\mathcal{L}$. Either t_1 is an initial segment of s_1 or s_1 is an initial segment of t_1. In either case Theorem 2.8 shows that t_1 and s_1 coincide. Next either t_2 is an initial segment of s_2 or s_2 is an initial segment of t_2. In either case Theorem 2.8 shows that t_2 and s_2 coincide. This proves **(2)**.

Now we prove **(3)**. We have five exclusive cases.

- γ is atomic.

- γ is $\bot$.
- γ starts with left parentheses. Suppose that
$$(\phi_1 \circ_1 \psi_1) \equiv \gamma \equiv (\phi_2 \circ_2 \psi_2)$$
where $\circ_1$, $\circ_2$ are propositional connectives $\rightarrow, \wedge, \vee$ and $\phi_1, \phi_2, \psi_1, \psi_2$ are formulas of $\mathcal{L}$. Then either ϕ_1 is initial segment of ϕ_2 or ϕ_2 is initial segment of ϕ_1. In either case by Theorem 2.8 we deduce that ϕ_1 coincides with ϕ_2. Hence $\circ_1$ coincides with $\circ_2$ and ψ_1 coincides with ψ_2.
- γ is $\forall_x \phi$ for a unique variable x and a unique formula ϕ of $\mathcal{L}$.
- γ is $\exists_x \phi$ for a unique variable x and a unique formula ϕ of $\mathcal{L}$.

This completes the proof. □

Definition 2.10. We define collection $\mathbf{Fr}(\gamma)$ of variables for every formula γ of $\mathcal{L}$. If γ is an atomic formula, then $\mathbf{Fr}(\gamma)$ consists of all variables occurring in γ. If γ is $\bot$, then $\mathbf{Fr}(\gamma)$ is empty. If γ is of the form
$$(\phi \rightarrow \psi), (\phi \wedge \psi), (\phi \vee \psi)$$
for some formulas ϕ, ψ of $\mathcal{L}$, then $\mathbf{Fr}(\gamma)$ is the union ot $\mathbf{Fr}(\phi)$ and $\mathbf{Fr}(\psi)$. Finally if γ is of the form
$$\forall x\, \phi,\ \exists_x \phi$$
for some formula ϕ of $\mathcal{L}$, then $\mathbf{Fr}(\gamma)$ is $\mathbf{Fr}(\phi)$ with x removed. The elements of $\mathbf{Fr}(\gamma)$ are *free variables* of γ.

Definition 2.11. Let σ be a formula of $\mathcal{L}$. If $\mathbf{Fr}(\sigma)$ is empty, then σ is a *sentence* of $\mathcal{L}$.

Now we describe central notion of substitutability and the procedure of substitution.

Definition 2.12. Let x be a variable and let t, s be terms of $\mathcal{L}$. Then $t[s/x]$ is the term of $\mathcal{L}$ obtained by replacing each occurrence of x in t by s. We say that $t[s/x]$ is obtained by *substitution* of x by s in t.

Definition 2.13. Let x be a variable and let t be a term of $\mathcal{L}$. We define *substitutability* of t for x in a formula γ of $\mathcal{L}$ as follows.

(1) If γ is an atomic formula, then t is substitutable for x in γ.

(2) If γ is $\bot$, then t is substitutable for x in γ.

(3) If γ has one of the forms
$$(\phi \rightarrow \psi), (\phi \wedge \psi), (\phi \vee \psi)$$
for some formulas ϕ, ψ of $\mathcal{L}$, then t is substitutable for x in γ if and only if t is substitutable for x in ϕ and t is substitutable for x in ψ.

(4) If γ has one of the forms
$$\forall_y \phi, \exists_y \phi$$
for a formula ϕ of $\mathcal{L}$ and a variable y distinct from x, then t is substitutable for x in γ if and only if either x does not occur free in ϕ or t is substitutable for x in ϕ and y does not occur in t.

(5) If γ has one of the forms
$$\forall_x \phi, \exists_x \phi$$
for a formula ϕ of $\mathcal{L}$, then t is substitutable for x in γ.

Definition 2.14. Let x be a variable and let t be a term of $\mathcal{L}$. Suppose that t is substitutable for x in γ. Then we define a formula $\gamma[t/x]$ of $\mathcal{L}$ as follows.

(1) If γ is an atomic formula, then $\gamma[t/x]$ is obtained from γ by replacing each occurrence of variable x by t.

(2) If γ is $\bot$, then $\gamma[t/x]$ is $\bot$.

(3) If γ has form

$$(\phi \to \psi), (\phi \wedge \psi), (\phi \vee \psi)$$

for some formulas ϕ, ψ of $\mathcal{L}$, then $\gamma[t/x]$ is

$$(\phi[t/x] \to \psi[t/x]), (\phi[t/x] \wedge \psi[t/x]), (\phi[t/x] \vee \psi[t/x])$$

(4) If γ has form

$$\forall_y \phi, \exists_y \phi$$

for a formula ϕ of $\mathcal{L}$ and a variable y distinct from x, then $\gamma[t/x]$ is

$$\forall_y \phi[t/x], \exists_y \phi[t/x]$$

(5) If γ has form

$$\forall_x \phi, \exists_x \phi$$

for a formula ϕ of $\mathcal{L}$, then $\gamma[t/x]$ coincides with γ.

We say that $\gamma[t/x]$ is obtained by *substitution* of x by t in γ.

3. Hilbert-style deductive systems for propositional extensions of minimal logic

Fix a first-order language $\mathcal{L}$. We describe a proof system for the collection of formulas of $\mathcal{L}$. This proof system serves as a basis for further developments of intuitionistic and classical logics.

Definition 3.1. The following are *quantifier axiom schemas* for first-order logic.

1. **Universal instantiation.**
$$\forall_x \phi \to \phi[t/x]$$
for every formula ϕ of $\mathcal{L}$ and every term t of $\mathcal{L}$ that is substitutable in ϕ for x.
2. **Universal generalization.**
$$\phi \to \forall_x \phi$$
for every formula ϕ of $\mathcal{L}$ and a variable x which is not free in ϕ.
3. **Distributivity of the universal quantifier.**
$$\forall_x(\phi \to \psi) \to (\forall_x \phi \to \forall_x \psi)$$
for all formulas ϕ, ψ of $\mathcal{L}$.
4. **Existential introduction.**
$$\phi[t/x] \to \exists_x \phi$$
for every formula ϕ of $\mathcal{L}$ and every term t of $\mathcal{L}$ that is substitutable in ϕ for x.
5. **Existential quantifier shift.**
$$\forall_x (\phi \to \psi) \to (\exists_x \phi \to \psi)$$
for all formulas ϕ, ψ of $\mathcal{L}$ such that x is not free in ψ.

Definition 3.2. The following are *propositional axiom schemas* of *minimal logic*.

1. **Implication schemas:**
$$\phi \to (\psi \to \phi)$$
$$(\phi \to (\psi \to \theta)) \to ((\phi \to \psi) \to (\phi \to \theta))$$
2. **Conjunction schemas:**

$$(\phi \wedge \psi) \to \phi$$
$$(\phi \wedge \psi) \to \psi$$
$$\phi \to (\psi \to (\phi \wedge \psi))$$

3. **Disjunction schemas:**

$$\phi \to (\phi \vee \psi)$$
$$\psi \to (\phi \vee \psi)$$
$$(\phi \to \theta) \to ((\psi \to \theta) \to ((\phi \vee \psi) \to \theta))$$

In the axiom schemas ϕ, ψ and θ denote arbitrary formulas of $\mathcal{L}$.

Definition 3.3. Let $\mathcal{L}$ be a first order language with identity. The *axioms of identity* of $\mathcal{L}$ are the following classes of formulas of $\mathcal{L}$.

1. $t = t$ for every term t of $\mathcal{L}$.
2. $(t_1 = t_2 \to t_2 = t_1)$ for all terms t_1, t_2 of $\mathcal{L}$.
3. $\big(t_1 = t_2 \to (t_2 = t_3 \to t_1 = t_3)\big)$ for all terms t_1, t_2, t_3 of $\mathcal{L}$.
4. $\underbrace{\big(t_1 = t_1' \to \big(t_2 = t_2' \to \big(\ldots \to (t_n = t_n')\ldots\big)\big)\big)}_{n \text{ symbols of equality}} \to \big(f t_1 \ldots t_n = f t_1' \ldots t_n'\big)$ where $t_1, \ldots, t_n$ and $t_1', \ldots, t_n'$ are terms of $\mathcal{L}$ and f is an n-ary function symbol of $\mathcal{L}$.
5. $\underbrace{\big(t_1 = t_1' \to \big(t_2 = t_2' \to \big(\ldots \to (t_n = t_n')\ldots\big)\big)\big)}_{n \text{ symbols of equality}} \to \big(r t_1 \ldots t_n \to r t_1' \ldots t_n'\big)$ where $t_1, \ldots, t_n$ and $t_1', \ldots, t_n'$ are terms of $\mathcal{L}$ and r is an n-ary relation symbol of $\mathcal{L}$.

Definition 3.4. Let ϕ be a formula of $\mathcal{L}$ and let $x_1, \ldots, x_n$ be variables. Then a formula

$$\underbrace{\forall_{x_1} \ldots \forall_{x_n}}_{n \text{ quantifiers}} \phi$$

is called a *generalization* of ϕ.

Definition 3.5. The axioms of minimal logic consist of:

1. all generalizations of quantifier axiom schemas,
2. all generalizations of propositional axiom schemas of minimal logic,
3. if $\mathcal{L}$ contains identity, all generalizations of the identity axioms.

Having defined the axioms we now introduce the notion of a deduction.

Definition 3.6. Let Γ be a collection of formulas of $\mathcal{L}$ and let ϕ be a formula of $\mathcal{L}$. A *deduction* of ϕ from Γ is a sequence

$$D_1, \ldots, D_n$$

of formulas of $\mathcal{L}$ such that D_n is ϕ, each D_k is either an axiom, an element of Γ or there exist $i, j < k$ such that D_j is of the form $(D_i \to D_k)$.

Remark 3.7. Let ϕ, ψ be formulas of $\mathcal{L}$. The rule that allows one to deduce ψ from ϕ and $(\phi \to \psi)$ is *modus ponens*.

Definition 3.8. Let Γ be a collection of formulas of $\mathcal{L}$ and let ϕ be a formula of $\mathcal{L}$. If there exists a deduction of ϕ from Γ, then ϕ is *deducible* from Γ. This is also denoted by $\Gamma \vdash \phi$.

Definition 3.9. Let ϕ be a formula of $\mathcal{L}$. If ϕ is deducible in $\mathcal{L}$ from the empty collection of formulas, then ϕ is a *theorem* of minimal logic.

Remark 3.10. Unless otherwise stated all results of the next sections will hold for Hilbert-style deductive systems that extend the system for minimal logic by introducing new propositional axiom schemas and their generalizations.

4. Basic Metatheorems

Fix a first-order language $\mathcal{L}$. In this section we prove basic results concerning the deductive systems introduced in the previous section. We start with elementary results.

Proposition 4.1 [monotonicity of $\vdash$]. *Let Γ, Δ be collections of formulas of $\mathcal{L}$ and let ϕ be a formula of $\mathcal{L}$. Suppose that $\Gamma \vdash \phi$ and $\Gamma \subseteq \Delta$. Then $\Delta \vdash \phi$.*

Proof. Every deduction of ϕ from Γ is also a deduction of ϕ from Δ. □

Proposition 4.2 [compactness of $\vdash$]. *Let Γ be a collection of formulas of $\mathcal{L}$ and let ϕ be a formula of $\mathcal{L}$. Suppose that $\Gamma \vdash \phi$. Then there exists finite subset Δ of Γ such that $\Delta \vdash \phi$.*

Proof. Every deduction of ϕ from Γ involves only finitely many formulas in Γ. □

Proposition 4.3 [transitivity of $\vdash$]. *Let Γ, Δ be collections of formulas of $\mathcal{L}$ and let ϕ be a formula of $\mathcal{L}$. Suppose that $\Gamma \vdash \psi$ for every formula $\psi \in \Delta$ and $\Delta \vdash \phi$. Then $\Gamma \vdash \phi$.*

Proof. By Proposition 4.2 there exist finitely many formulas $\psi_1, ..., \psi_m$ of Δ such that

$$\psi_1, ..., \psi_m \vdash \phi$$

Now for each i there exists a deduction of ψ_i from Γ. We concatenate these deductions for $i = 1, ..., m$ and further concatenate them with a deduction of ϕ from $\psi_1, ..., \psi_m$. This yields a deduction of ϕ from Γ. □

Fact 4.4. *Let ϕ be a formula of $\mathcal{L}$. Then $\vdash \phi \to \phi$ is a theorem of minimal logic.*

Proof. Consider the following sequence of formulas.

1. $\phi \to (\phi \to \phi)$
2. $\phi \to ((\phi \to \phi) \to \phi)$
3. $(\phi \to ((\phi \to \phi) \to \phi)) \to ((\phi \to (\phi \to \phi)) \to (\phi \to \phi))$
4. $(\phi \to (\phi \to \phi)) \to (\phi \to \phi)$
5. $\phi \to \phi$

Note that **1**,**2**,**3** are logical axioms. Next **4** is obtained by modus ponens from **2**,**3**. Finally, **5** is also derived by modus ponens from **1**,**4**. Thus the sequence above is a deduction of $\phi \to \phi$ from empty collection of formulas. □

Theorem 4.5 [deduction theorem]. *Let Γ be a collection of formulas of $\mathcal{L}$ and let ϕ, ψ be formulas of $\mathcal{L}$. Then $\Gamma, \phi \vdash \psi$ if and only if $\Gamma \vdash \phi \to \psi$.*

Proof. Note that

$$\phi, \phi \to \psi, \psi$$

is a deduction of ψ from $\phi, \phi \to \psi$. If $\Gamma \vdash \phi \to \psi$, then by Proposition 4.3 we derive that $\Gamma, \phi \vdash \psi$.

Now suppose that $\Gamma, \phi \vdash \psi$. Let

$$D_1, ..., D_n$$

be a deduction of ψ from Γ, ϕ. We prove that $\Gamma \vdash \phi \to D_k$ for each k. The proof goes on induction on k. We consider the following cases.

(1) D_k is an axiom of $\mathcal{L}$ or an element of Γ. Note that

$$D_k \to (\phi \to D_k)$$

is a propositional axiom of $\mathcal{L}$. Hence

$$D_k, D_k \to (\phi \to D_k), \phi \to D_k$$

is a deduction of $\phi \to D_k$ from Γ.

(2) D_k is ϕ. Then $(\phi \to D_k)$ is a theorem of minimal logic by Fact 4.4 and hence it is deducible from Γ.

(3) There exist $i, j < k$ such that D_j is $(D_i \to D_k)$. Note that

$$\big(\phi \to (D_i \to D_k)\big) \to \big((\phi \to D_i) \to (\phi \to D_k)\big)$$

is an axiom of $\mathcal{L}$ and hence one can write a deduction of $\phi \to D_k$ from

$$\phi \to D_i, \phi \to D_j$$

By induction $\Gamma \vdash \phi \to D_i$ and $\Gamma \vdash \phi \to D_j$. Thus also $\Gamma \vdash \phi \to D_k$ according to Proposition 4.3.

This proves that $\Gamma \vdash \phi \to D_k$ for all k. Hence $\Gamma \vdash \phi \to D_n$. Since D_n is ψ, we derive that $\Gamma \vdash \phi \to \psi$. □

Theorem 4.6 [variable generalization]. *Let Γ be a collection of formulas of $\mathcal{L}$ and let x be a variable which does not occur as a free variable in any formula in Γ. If $\Gamma \vdash \phi$ for some formula ϕ of $\mathcal{L}$, then $\Gamma \vdash \forall_x \phi$.*

Proof. Let

$$D_1, ..., D_n$$

be a deduction of ϕ from Γ. We prove that for each k there exists a deduction of $\forall_x D_k$ from Γ. For this we consider the following cases.

(1) D_k is an axiom. Since axioms are closed under generalizations, we derive that $\forall_x D_k$ is also an axiom. Hence $\Gamma \vdash \forall_x D_k$.

(2) D_k is an element of Γ. Then

$$D_k, D_k \to \forall_x D_k, \forall_x D_k$$

is a deduction of $\forall_x D_k$ from Γ. Note that here we use the assumption that there are no formulas in Γ in which x occurs as a free variable. Indeed, from this assumption it follows that $D_k \to \forall_x D_k$ is an instance of universal generalization schema.

(3) There exist $i, j < k$ such that D_j is $D_i \to D_k$. Consider the sequence of formulas

$$\forall_x D_i, \forall_x (D_i \to D_k), \forall_x (D_i \to D_k) \to (\forall_x D_i \to \forall_x D_k), \forall_x D_i \to \forall_x D_k, \forall_x D_k$$

If there are deductions of $\forall_x D_i$ and $\forall_x D_j$ that is $\forall_x (D_i \to D_k)$ from Γ, then we add the sequence above to their concatenation in order to obtain a deduction of $\forall_x D_k$ from Γ.

From **(1)**, **(2)** and **(3)** we derive that the assertion follows by induction on k. In particular, there exists a deduction of $\forall_x D_n$ that is $\forall_x \phi$ from Γ. □

Now we apply the above meta-theorems to present some theorems of minimal logic.

Proposition 4.7 [α-equivalence]. *Let ϕ be a formula of $\mathcal{L}$ and let y be a variable. Suppose that y is substitutable for x in ϕ and is not free in ϕ. Then*

$$\forall_x \phi \leftrightarrow \forall_y \phi[y/x],\ \exists_x \phi \leftrightarrow \exists_y \phi[y/x]$$

are theorems of minimal logic.

Proof. Note that

$$\forall_x \phi, \forall_x \phi \rightarrow \phi[y/x], \phi[y/x]$$

is a deduction of $\phi[y/x]$ from $\forall_x \phi$ in $\mathcal{L}$. By Theorem 4.6 we infer that $\forall_x \phi \vdash \forall_y \phi[y/x]$ and by Theorem 4.5 we obtain $\vdash \forall_x \phi \rightarrow \forall_y \phi[y/x]$. Since x is substitutable for y in $\phi[y/x]$ and $\phi \equiv \phi[y/x][x/y]$, there is symmetry in x and y and hence

$$\vdash \forall_y \phi[y/x] \rightarrow \forall_x \phi$$

This proves that the first formula is a theorem of minimal logic.

Since $\phi \rightarrow \exists_y \phi[y/x]$ is an instance of existential introduction schema, we derive that

$$\forall_x \left(\phi \rightarrow \exists_y \phi[y/x]\right)$$

is an axiom of minimal logic. Applying existential quantifier shifting combined with modus ponens implies that

$$\vdash \exists_x \phi \rightarrow \exists_y \phi[y/x]$$

The other direction follows by symmetry between y and x. Hence the second formula is a theorem of minimal logic. □

Proposition 4.8. *Let ϕ be a formula of $\mathcal{L}$ and let x, y be variables. Then*

$$\forall_x \forall_y \phi \leftrightarrow \forall_y \forall_x \phi,\ \exists_x \exists_y \phi \leftrightarrow \exists_y \exists_x \phi$$

are theorems of minimal logic.

Proof. Since $\forall_y \phi \rightarrow \phi$ is an instance of universal instantiation, we derive that

$$\forall_x \left(\forall_y \phi \rightarrow \phi\right)$$

is an axiom of minimal logic. Thus by distributivity of universal quantifier we infer

$$\vdash \forall_x \forall_y \phi \rightarrow \forall_x \phi$$

By Theorem 4.5 and Theorem 4.6 we deduce that $\forall_x \forall_y \phi \vdash \forall_y \forall_x \phi$. We apply Theorem 4.5 to obtain

$$\vdash \forall_x \forall_y \phi \rightarrow \forall_y \forall_x \phi$$

Now the fact that the first formula is a theorem of minimal logic follows from symmetry between x and y.

By instances of existential introduction $\phi \rightarrow \exists_x \phi$ and $\exists_x \phi \rightarrow \exists_y \exists_x \phi$. Hence

$$\vdash \phi \rightarrow \exists_y \exists_x \phi$$

Theorem 4.6 implies that

$$\vdash \forall_x \forall_y \left(\phi \rightarrow \exists_y \exists_x \phi\right)$$

Next we apply existential quantifier shift twice to derive

$$\vdash \exists_x \exists_y \phi \leftrightarrow \exists_y \exists_x \phi$$

Now the fact that the second formula is a theorem of minimal logic follows from symmetry of x and y. □

5. Conservativity of definitional extensions by relational symbols

In this section we prove a meta-theoretical result that states that an extension of the original language by names for certain formulas is conservative with respect to the deductive system. First we need to rigorously define the concept of conservative extension.

Let $\mathcal{L}$ be a first-order language and let Γ be a collection of formulas of $\mathcal{L}$. If a formula ϕ of $\mathcal{L}$ is deducible from Γ, then we write $\Gamma \vdash_{\mathcal{L}} \phi$. This makes explicit that deductions consist of sequences of formulas of $\mathcal{L}$.

Definition 5.1. Let $\mathcal{L}$ and $\mathcal{M}$ be first-order languages. Suppose that the signature of $\mathcal{L}$ is contained in that of $\mathcal{M}$. Then $\mathcal{M}$ is an *extension* of $\mathcal{L}$.

Definition 5.2. Let $\mathcal{L}$ be a first-order language and let $\mathcal{M}$ be its extension. Let Γ be a collection of formulas of $\mathcal{L}$. Suppose that Δ is a collection of formulas of $\mathcal{M}$ that contains Γ. Then Δ is an *extension* of Γ.

Definition 5.3. Let $\mathcal{L}$ be a first-order language and let $\mathcal{M}$ be its extension. Let Γ be a collection of formulas of $\mathcal{L}$ and let Δ be a collection of formulas of $\mathcal{M}$ extending Γ. Suppose that for every formula ϕ of $\mathcal{L}$ if $\Delta \vdash_{\mathcal{M}} \phi$, then $\Gamma \vdash_{\mathcal{L}} \phi$. Then Δ is a *conservative* extension of Γ.

Theorem 5.4. *Let $\mathcal{L}$ be a first-order language and let $\mathcal{R}$ be a family of formulas of $\mathcal{L}$. Consider a first-order language $\mathcal{L}[\mathcal{R}]$ obtained from $\mathcal{L}$ by adding a fresh distinct relational symbol r_θ for every formula θ in $\mathcal{R}$. Moreover, the arity of r_θ is the number of free variables occurring in θ.*

Let Γ be a collection of formulas of $\mathcal{L}$. Let $\Gamma[\mathcal{R}]$ be a collection of formulas of $\mathcal{L}[\mathcal{R}]$ obtained from Γ by adding for each θ in $\mathcal{R}$ a formula

$$\forall_{x_1} ... \forall_{x_k} (\theta \leftrightarrow r_\theta x_1 ..., x_k)$$

where $x_1, ..., x_k$ are all variables that occur as free in θ.

Then $\Gamma[\mathcal{R}]$ is a conservative extension of Γ.

For the proof we need the following result.

Lemma 5.4.1. *Let $\mathcal{L}$ be a first-order language and let V be a finite collection of variables of $\mathcal{L}$. Let θ be a formula of $\mathcal{L}$. Then there exists a formula $\alpha(\theta)$ of $\mathcal{L}$ such that the following assertions hold.*

- **(1)** *$\alpha(\theta) \leftrightarrow \theta$ is a theorem of minimal logic.*
- **(2)** *The collection of bound variables of $\alpha(\theta)$ is disjoint from V.*
- **(3)** *θ and $\alpha(\theta)$ have the same collection of free variables.*

Proof of the lemma. The construction of $\alpha(\theta)$ and the proof that it satisfies the assertions proceed by induction on the complexity of θ.

- $\alpha(\bot)$ is $\bot$. The assertions holds in that case.
- If θ is atomic, then we set $\alpha(\theta)$ to be θ. The assertions are straightforward.
- Suppose that θ has one of the forms

  $$(\theta_1 \to \theta_2),\ (\theta_1 \wedge \theta_2),\ (\theta_1 \vee \theta_2)$$

  Then $\alpha(\theta)$ is of the form

  $$(\alpha(\theta_1) \to \alpha(\theta_2)),\ (\alpha(\theta_1) \wedge \alpha(\theta_2)),\ (\alpha(\theta_1) \vee \alpha(\theta_2))$$

  respectively. If $\alpha(\theta_1)$ and $\alpha(\theta_2)$ satisfy the assertions, then $\alpha(\theta)$ also satisfies the assertions. Hence this case easily follows by induction.

- If θ is of the form $\forall_x \eta$ for some variable x and formula $\mathcal{L}$, then we pick a variable $y \notin V$ that does not occur in $\alpha(\eta)$ and we set

$$\alpha(\theta) \equiv \forall_y \alpha(\eta)[y/x]$$

Note that

$$\vdash_{\mathcal{L}} \forall_x \eta \leftrightarrow \forall_x \alpha(\eta)$$

by the induction hypothesis and

$$\forall_x \alpha(\eta) \leftrightarrow \forall_y \alpha(\eta)[y/x]$$

is a theorem of minimal logic by Propositions 4.7. Hence combining these two facts we infer

$$\vdash_{\mathcal{L}} \forall_x \eta \leftrightarrow \forall_y \alpha(\eta)[y/x]$$

This proves that **(1)** holds. The assertions **(2)** and **(3)** are easy consequences of induction hypothesis.

□

Lemma 5.4.2. *Let $\mathcal{L}$ be a first-order language and let θ be a formula of $\mathcal{L}$. Suppose that $x_1, ..., x_l$ are all variables that occur as free in θ. Consider a first-order language $\mathcal{L}[r]$ obtained from $\mathcal{L}$ by adding a fresh l-ary relational symbol r.*

Let Γ be a collection of formulas of $\mathcal{L}$. Consider a collection $\Gamma[r]$ of formulas of $\mathcal{L}[r]$ obtained from Γ by adding a formula

$$\forall_{x_1} ... \forall_{x_l} (\theta \leftrightarrow r x_1 ..., x_l)$$

Then $\Gamma[r]$ is a conservative extension of Γ.

Proof of the lemma. Let ϕ be a formula of $\mathcal{L}$ and suppose that $\Gamma[r] \vdash_{\mathcal{L}[r]} \phi$. Let

$$D_1, ..., D_n$$

be a deduction of ϕ from $\Gamma[r]$ in $\mathcal{L}[r]$. Let V be the collection of all variables occurring in formulas $D_1, ..., D_n$.

Let $\alpha(\theta)$ be a formula of $\mathcal{L}$ obtained from θ by means of Lemma 5.4.1.

Next suppose that $\mathfrak{U}$ is a collection of all subformulas occurring in $D_1, ..., D_n$. For each $\mu \in \mathfrak{U}$ we construct a formula μ^* of $\mathcal{L}$. The construction proceeds by induction on the complexity of μ.

(1) Suppose that μ is $\bot$, then μ^* is $\bot$.

(2) Suppose that μ is an atomic formula in $\mathcal{L}$, then μ^* is μ.

(3) Suppose that μ is of the form $rt_1 ... rt_l$ where r is the newly introduced relation symbol and $t_1, ..., t_l$ are terms of $\mathcal{L}$. Then

$$\mu^* \equiv \alpha(\theta)[t_1/x_1, ..., t_l/x_l]$$

Note that there is no issue with substitution because all variables occurring in $t_1, ..., t_l$ are in V and hence they are not bound by any quantifier in $\alpha(\theta)$.

(4) If μ has one of the forms

$$(\mu_1 \to \mu_2), (\mu_1 \wedge \mu_2), (\mu_1 \vee \mu_2)$$

for some formulas μ_1, μ_2, then μ^* is

$$(\mu_1^* \to \mu_2^*), (\mu_1^* \wedge \mu_2^*), (\mu_1^* \vee \mu_2^*)$$

respectively.

(5) If μ has one of the forms

$$\forall_x \eta, \exists_x \eta$$

for some formula ν, then μ^* is

$$\forall_x \eta^*, \exists_x \eta^*$$

respectively.

Observe that if μ is in $\mathcal{L}$, then μ^* and μ coincide.

Now we show that $D_1^*, ..., D_n^*$ gives rise to a deduction of ϕ from Γ.

Suppose that D_k is any axiom of $\mathcal{L}[r]$. Since the translation $\mu \mapsto \mu^*$ preserves connectives, quantifiers and substitution, we derive that D_k^* is an axiom of $\mathcal{L}$.

Note also that

$$\left(\forall_{x_1}...\forall_{x_l}\,(\theta \leftrightarrow rx_1...x_l)\right)^* \equiv \forall_{x_1}...\forall_{x_l}\,(\theta \leftrightarrow \alpha(\theta))$$

is a theorem of minimal logic by Lemma 5.4.1.

Finally, if $D_j \equiv (D_i \to D_k)$ for some $i, j < k$, then $D_j^* \equiv (D_i^* \to D_k^*)$.

Since $\mu \mapsto \mu^*$ preserves the formulas of $\mathcal{L}$, it follows that $D_1^*, ..., D_n^*$ is a deduction of ϕ from Γ in $\mathcal{L}$. Thus $\Gamma[r]$ is conservative extension of Γ. □

Proof of the theorem. Suppose that $\Gamma[\mathcal{R}] \vdash_{\mathcal{L}[\mathcal{R}]} \phi$ for some formula ϕ of $\mathcal{L}$. Since in any deduction of ϕ from $\Gamma[\mathcal{R}]$ in $\mathcal{L}[\mathcal{R}]$ only finitely many newly introduced relation symbols are used, we may assume that $\Gamma[r_{\theta_1}, ..., r_{\theta_n}] \vdash_{\mathcal{L}[r_{\theta_1},...,r_{\theta_n}]} \phi$ where $\theta_1, ..., \theta_n$ are formulas in $\mathcal{R}$. By repeated application of Lemma 5.4.2 we conclude $\Gamma \vdash_{\mathcal{L}} \phi$. This completes the proof. □

6. Constant elimination and indefinite descriptions of individuals

Let c be a constant and let ϕ be a formula of the same first-order language. For a variable x we denote by $\phi[x/c]$ the formula obtained by replacing each occurrence of c in ϕ by x.

Theorem 6.1 [constant elimination]. *Let Γ be a collection of formulas of a first-order language $\mathcal{L}$. Consider a first-order language $\mathcal{L}[c]$ obtained by adding a fresh constant c to $\mathcal{L}$. Suppose that $\Gamma \vdash_{\mathcal{L}[c]} \phi$ for some formula ϕ of $\mathcal{L}[c]$. Then $\Gamma \vdash_{\mathcal{L}} \forall_x \phi[x/c]$ for every variable x that does not occur in ϕ.*

Proof. By Proposition 4.2 there exists a finite subcollection Ξ of Γ such that $\Xi \vdash \phi$. Suppose that

$$D_1, ..., D_n$$

is a deduction of ϕ in $\mathcal{L}[c]$ from Ξ. Suppose that z is a variable which does not occur in any formula of Ξ and in any formula among $D_1, ..., D_n$. Since c does not occur in any formula in Ξ, we infer that

$$D_1[z/c], ..., D_n[z/c]$$

is a deduction of $\phi[z/c]$ in $\mathcal{L}$ from Ξ. By Theorem 4.6 we derive that $\Xi \vdash_{\mathcal{L}} \forall_z \phi[z/c]$. Proposition 4.1 implies that $\Gamma \vdash_{\mathcal{L}} \forall_z \phi[z/c]$.

Fix a variable x which does not occur in ϕ. Then x is substitutable in $\phi[z/c]$ for z and

$$\phi[z/c][x/z] \equiv \phi[x/c]$$

Thus

$$\forall_z \phi[z/c],\ \forall_z \phi[z/c] \to \phi[x/c],\ \phi[x/c]$$

is a deduction of $\phi[x/c]$ from $\forall_z \phi[z/c]$ in $\mathcal{L}$. Since x does not occur in $\phi[z/c]$, Theorem 4.6 implies that

$$\forall_z \phi[z/c] \vdash_{\mathcal{L}} \forall_x \phi[x/c]$$

Hence by Proposition 4.3 we obtain $\Gamma \vdash_{\mathcal{L}} \forall_x \phi[x/c]$. □

Next we present some consequences of constant elimination. The next result is a formalization of Russell's theory of indefinite descriptions.

Theorem 6.2. *Let Γ be a collection of formulas of a first-order language $\mathcal{L}$. Let $\mathcal{C}$ be a collection of formulas of $\mathcal{L}$ such that $\Gamma \vdash_{\mathcal{L}} \exists_{x_\theta} \theta$ for every θ in $\mathcal{C}$ and some variable x_θ of $\mathcal{L}$.*

Consider a first-order language $\mathcal{L}[\mathcal{C}]$ obtained from $\mathcal{L}$ by adding a fresh constants c_θ to $\mathcal{L}$ for each θ in $\mathcal{C}$. Let $\Gamma[\mathcal{C}]$ be a collection of formulas of $\mathcal{L}[\mathcal{C}]$ obtained from Γ by adding $\theta[c_\theta/x_\theta]$ for each θ in $\mathcal{C}$.

Then $\Gamma[\mathcal{C}]$ is a conservative extension of Γ.

The proof is based on the following result.

Lemma 6.2.1. *Let Γ be a collection of formulas of a first-order language $\mathcal{L}$. Let θ be a formula of $\mathcal{L}$ and let x be a variable such that $\Gamma \vdash_{\mathcal{L}} \exists_x \theta$.*

Consider a first-order language $\mathcal{L}[c]$ obtained from $\mathcal{L}$ by adding a fresh constant c. Let $\Gamma[c]$ be a collection of formulas of $\mathcal{L}[c]$ obtained from Γ by adding a formula $\theta[c/x]$.

Then $\Gamma[c]$ is a conservative extension of Γ.

Proof of the lemma. Suppose that ϕ is a formula of $\mathcal{L}$ such that $\Gamma[c] \vdash_{\mathcal{L}[c]} \phi$. According to the deduction theorem we have

$$\Gamma \vdash_{\mathcal{L}[c]} \theta[c/x] \to \phi$$

Fix a variable y which does not occur in $\theta[c/x] \to \phi$ and in θ. Theorem 6.1 together with the fact that ϕ is a formula of $\mathcal{L}$ implies that

$$\Gamma \vdash_{\mathcal{L}} \forall_y (\theta[c/x][y/c] \to \phi)$$

Note that $\theta[c/x][y/c] \equiv \theta[y/x]$. Thus $\Gamma \vdash_{\mathcal{L}} \forall_y (\theta[y/x] \to \phi)$. Since

$$\forall_y (\theta[y/x] \to \phi) \to \exists_y \theta[y/x] \to \phi$$

is an instance of existential quantifier shift, we infer that $\Gamma \vdash_{\mathcal{L}} \exists_y \theta[y/x] \to \phi$.

The formula $\theta \to \exists_y \theta[y/x]$ is an instance of existential introduction. Hence $\forall_x (\theta \to \exists_y \theta[y/x])$ is an instance of axiom schema. According to another instance of existential quantifier shift we derive that

$$\vdash_{\mathcal{L}} \exists_x \theta \to \exists_y \theta[y/x]$$

Since $\Gamma \vdash_{\mathcal{L}} \exists_x \theta$, we deduce that $\Gamma \vdash_{\mathcal{L}} \exists_y \theta[y/x]$.

Combining

$$\Gamma \vdash_{\mathcal{L}} \exists_y \theta[y/x]$$

with

$$\Gamma \vdash_{\mathcal{L}} \exists_y \theta[y/x] \to \phi$$

we infer $\Gamma \vdash_{\mathcal{L}} \phi$. This proves that $\Gamma[c]$ is conservative extension of Γ. □

Proof of the theorem. Suppose that $\Gamma[\mathcal{C}] \vdash_{\mathcal{L}[\mathcal{C}]} \phi$ for some formula ϕ of $\mathcal{L}$. Since in any deduction of ϕ from $\Gamma[\mathcal{C}]$ in $\mathcal{L}[\mathcal{C}]$ only finitely many newly introduced constant symbols are used, we may assume that $\Gamma[c_{\theta_1}, ..., c_{\theta_n}] \vdash_{\mathcal{L}[c_{\theta_1}, ..., c_{\theta_n}]} \phi$ where $\theta_1, ..., \theta_n$ are formulas in $\mathcal{C}$. By repeated application of Lemma 6.2.1 we conclude $\Gamma \vdash_{\mathcal{L}} \phi$. This completes the proof. □

Now we present some applications of the theorem.

Remark 6.3. Let $\mathcal{L}$ be a first-order language and let ϕ be a formula of $\mathcal{L}$. Suppose that y is a variable and that y_1, y_2 are variables substitutable for y in ϕ. Then we introduce the following shorthand

$$\exists!_y \phi \equiv \exists_y \phi \wedge \forall_{y_1} \forall_{y_2} ((\phi[y_1/y] \wedge \phi[y_2/y]) \to y_1 = y_2)$$

Proposition 6.4. *Let $\mathcal{L}$ be a first-order language and let ϕ, ψ, ρ be formulas of $\mathcal{L}$. Then the following assertions hold.*

(1) $\exists!_y \phi \vdash_{\mathcal{L}} \exists_y ((\psi \vee \rho) \wedge \phi) \leftrightarrow \big(\exists_y(\psi \wedge \phi) \vee \exists_y(\rho \wedge \phi)\big)$

(2) $\exists!_y \phi \vdash_{\mathcal{L}} \exists_y ((\psi \wedge \rho) \wedge \phi) \leftrightarrow \big(\exists_y(\psi \wedge \phi) \wedge \exists_y(\rho \wedge \phi)\big)$

(3) $\exists!_y \phi \vdash_{\mathcal{L}} \exists_y ((\psi \to \rho) \wedge \phi) \leftrightarrow \big(\exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi)\big)$

Proof. We leave the proofs of **(1)** and **(2)** for the reader.

We prove the more difficult direction of the assertion **(3)** and leave the other part for the reader.

First we introduce two fresh constant symbols c, d and extend $\mathcal{L}$ to $\mathcal{L}[c]$ and $\mathcal{L}[c][d]$. We have

$$\exists!_y \phi, \phi[c/y], \phi[d/y] \vdash_{\mathcal{L}[c][d]} c = d$$

Thus we have

$$\exists!_y \phi, \exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi), \phi[c/y], \psi[c/y], \rho[d/y] \wedge \phi[d/y] \vdash_{\mathcal{L}[c][d]} \rho[c/y]$$

Observe that

$$\exists!_y \phi, \exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi), \phi[c/y], \psi[c/y] \vdash_{\mathcal{L}[c]} \exists_y(\rho \wedge \phi)$$

We combine these two results with Theorem 6.2 to infer that

$$\exists!_y \phi, \exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi), \phi[c/y], \psi[c/y] \vdash_{\mathcal{L}[c]} \rho[c/y]$$

Thus

$$\exists!_y \phi, \exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \psi[c/y] \to \rho[c/y]$$

by deduction theorem and derive

$$\exists!_y \phi, \exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \exists_y ((\psi \to \rho) \wedge \phi)$$

Since $\exists!_y \phi \vdash_{\mathcal{L}} \exists_y \phi$, we apply Theorem 6.2 to derive that

$$\exists!_y \phi, \exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi) \vdash_{\mathcal{L}} \exists_y ((\psi \to \rho) \wedge \phi)$$

Finally, by deduction theorem we obtain

$$\exists!_y \phi \vdash_{\mathcal{L}} \big(\exists_y(\psi \wedge \phi) \to \exists_y(\rho \wedge \phi)\big) \to \exists_y ((\psi \to \rho) \wedge \phi)$$

□

Proposition 6.5. *Let $\mathcal{L}$ be a first-order language and let ϕ, ψ be formulas of $\mathcal{L}$. Let x be a variable that is not free in ϕ. Then*

$$\exists!_y \phi \vdash_{\mathcal{L}} \forall_x \exists_y(\psi \wedge \phi) \leftrightarrow \exists_y \forall_x(\psi \wedge \phi)$$

Proof. We prove the more difficult part of the assertion and leave the other one for the reader.

First we introduce two fresh constant symbols c, d and extend $\mathcal{L}$ to $\mathcal{L}[c]$ and $\mathcal{L}[c][d]$. We have

$$\exists!_y \phi, \phi[c/y], \psi[d/y] \wedge \phi[d/y] \vdash_{\mathcal{L}[c][d]} c = d$$

Thus

$$\exists!_y \phi, \phi[c/y], \psi[d/y] \wedge \phi[d/y] \vdash_{\mathcal{L}[c][d]} \psi[c/y]$$

Since

$$\exists!_y \phi, \forall_x \exists_y(\psi \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \exists_y(\psi \wedge \phi)$$

by universal instantiation, we derive according to Theorem 6.2 that

$$\exists!_y \phi, \forall_x \exists_y(\psi \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \psi[c/y]$$

Therefore, we obtain

$$\exists!_y \phi, \forall_x \exists_y(\psi \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \psi[c/y] \wedge \phi[c/y]$$

Since x is not free in ϕ, variable generalization implies that

$$\exists!_y\, \phi, \forall_x \exists_y (\psi \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \forall_x (\psi[c/y] \wedge \phi[c/y])$$

Next we use existential introduction to infer

$$\exists!_y\, \phi, \forall_x \exists_y (\psi \wedge \phi), \phi[c/y] \vdash_{\mathcal{L}[c]} \exists_y \forall_x (\psi \wedge \phi)$$

Again by Theorem 6.2 we infer that

$$\exists!_y\, \phi, \forall_x \exists_y (\psi \wedge \phi) \vdash_{\mathcal{L}} \exists_y \forall_x (\psi \wedge \phi)$$

and hence by deduction theorem

$$\exists!_y\, \phi \vdash_{\mathcal{L}} \forall_x \exists_y (\psi \wedge \phi) \rightarrow \exists_y \forall_x (\psi \wedge \phi)$$

□

7. Conservativity of definite descriptions

The goal of this section is to formalize a generalization of Russell's theory of definite descriptions.

Theorem 7.1. *Let Γ be a collection of formulas of a first-order language $\mathcal{L}$. Let $\mathcal{F}$ be a family of formulas of $\mathcal{L}$. Suppose that for every θ in $\mathcal{F}$ there exists variable y_θ that occurs free in θ such that*

$$\Gamma \vdash_{\mathcal{L}} \forall_{x_1} ... \forall_{x_l} \exists!_{y_\theta}\, \theta$$

where $x_1, ..., x_l$ are the remaining free variables in θ.

Consider a first-order language $\mathcal{L}[\mathcal{F}]$ obtained from $\mathcal{L}$ by adding a fresh distinct l-ary function symbol f_θ for each θ in $\mathcal{L}$. Let $\Gamma[\mathcal{F}]$ be a collection of formulas of $\mathcal{L}[\mathcal{F}]$ obtained from Γ by adding for each θ in $\mathcal{F}$ a formula

$$\forall_{x_1} ... \forall_{x_l} (\theta \leftrightarrow y_\theta = f x_1 ... x_l)$$

Then $\Gamma[\mathcal{F}]$ is a conservative extension of Γ.

For the proof we need the following result.

Lemma 7.1.1. *Let $\mathcal{L}$ be a first-order language and let Γ be a collection of formulas of $\mathcal{L}$. Let F be a $(l+1)$-ary relation symbol of $\mathcal{L}$ for some natural number l. Suppose that $x_1, ..., x_l, y_f$ are variables of $\mathcal{L}$ such that*

$$\Gamma \vdash_{\mathcal{L}} \forall_{x_1} ... \forall_{x_l} \exists!_{y_f}\, F x_1 ... x_l y_f$$

Let $\mathcal{L}[f]$ be the first-order language obtained from $\mathcal{L}$ by adding a fresh l-ary function symbol f and let $\Gamma[f]$ be a collection of formulas of $\mathcal{L}[f]$ obtained from Γ by adding

$$\forall_{x_1} ... \forall_{x_l}\, F x_1 ... x_l f x_1 ... x_l$$

Then $\Gamma[f]$ is a conservative extension of Γ.

Proof of the lemma. Suppose that ϕ is a formula of $\mathcal{L}$ such that $\Gamma[f] \vdash_{\mathcal{L}[f]} \phi$. Let

$$D_1, ..., D_m$$

be a deduction of ϕ from $\Gamma[f]$ in $\mathcal{L}[f]$. Let V be the collection of all variables occurring in formulas $D_1, ..., D_m$.

For each term t occurring in $D_1, ..., D_m$ we choose a unique variable y_t distinct from all variables in V. Next suppose that $\mathfrak{U}$ is a collection of all terms and formulas occurring as subexpressions in $D_1, ..., D_m$. For each $\mu \in \mathfrak{U}$ we construct a formula μ^* of $\mathcal{L}$. The construction proceeds by induction on the complexity of μ.

(1) Suppose that μ is a variable x, then μ^* is $y_\mu = x$.

(2) Suppose that μ is a term of the form $g\mu_1...\mu_n$ where g is an n-ary function symbol in $\mathcal{L}$ and $\mu_1, ..., \mu_n$ are terms. Then

$$\mu^* \equiv \exists_{y_{\mu_1}} ... \exists_{y_{\mu_n}} \left(y_\mu = g y_{\mu_1} ... y_{\mu_n} \wedge \mu_1^* \wedge ... \wedge \mu_n^* \right)$$

(3) Suppose that μ is a term of the form $f\mu_1...\mu_l$ where $\mu_1, ..., \mu_l$ are terms. Then

$$\mu^* \equiv \exists_{y_{\mu_1}} ... \exists_{y_{\mu_l}} \left(F y_{\mu_1} ... y_{\mu_l} y_\mu \wedge \mu_1^* \wedge ... \wedge \mu_l^* \right)$$

(4) Suppose that μ is $\bot$, then μ^* is $\bot$.

(5) Suppose that μ is an atomic formula of the form $r\mu_1...\mu_n$ where r is an n-ary relation symbol and $\mu_1, ...\mu_n$ are terms. Then

$$\mu^* \equiv \exists_{y_{\mu_1}} ... \exists_{y_{\mu_n}} \left(r y_{\mu_1} ... y_{\mu_n} \wedge \mu_1^* \wedge ... \wedge \mu_n^* \right)$$

(6) If μ has one of the forms

$$(\mu_1 \to \mu_2), (\mu_1 \wedge \mu_2), (\mu_1 \vee \mu_2)$$

for some formulas μ_1, μ_2, then μ^* is

$$(\mu_1^* \to \mu_2^*), (\mu_1^* \wedge \mu_2^*), (\mu_1^* \vee \mu_2^*)$$

respectively.

(7) If μ has one of the forms

$$\forall_x \eta, \exists_x \eta$$

for some formula θ, then μ^* is

$$\forall_x \eta^*, \exists_x \eta^*$$

respectively.

The following assertions hold for every μ in $\mathfrak{U}$.

Fact 1. If μ is a term and $x_1, ..., x_n$ are all variables that occur in μ, then $x_1, ..., x_n, y_\mu$ are all free variables occurring in μ^*.

Fact 2. If μ is a term of $\mathcal{L}$, then $\vdash_{\mathcal{L}} \mu^* \leftrightarrow y_\mu = \mu$.

Fact 3. If μ is a term, then μ^* is a functional formula in Γ with respect to y_μ.

Fact 4. If μ is a formula of $\mathcal{L}$, then $\vdash_{\mathcal{L}} \mu^* \leftrightarrow \mu$.

Fact 5. Suppose that μ is a term. If s is a term in $\mathfrak{U}$ and x is a variable such that $\mu[s/x]$ is in $\mathfrak{U}$, then

$$\Gamma \vdash_{\mathcal{L}} \exists_y \left(\mu^*[y_{\mu[s/x]}/y_\mu, y/x] \wedge s^*[y/y_s] \right) \leftrightarrow \mu[s/x]^*$$

for every variable y not in V and distinct from y_t for all terms t in $\mathfrak{U}$.

Fact 6. Suppose that μ is a formula. If s is a term in $\mathfrak{U}$ and x is a variable such that s is substitutable in μ for x and $\mu[s/x]$ is in $\mathfrak{U}$, then

$$\Gamma \vdash_{\mathcal{L}} \exists_y \left(\mu^*[y/x] \wedge s^*[y/y_s] \right) \leftrightarrow \mu[s/x]^*$$

for every variable y not in V and distinct from y_t for all terms t in $\mathfrak{U}$.

Proofs proceed by induction on the complexity of μ. We leave (easier) proofs of Facts 1-4 for the reader. We prove Facts 5-6 together since they are closely related.

(1) If μ is a variable x, then we need to prove that

$$\Gamma \vdash_{\mathcal{L}} \exists_y \left(y_s = y \wedge s^*[y/y_s] \right) \leftrightarrow s^*$$

The right hand side is a theorem of minimal logic. In particular, it can be deduced from Γ in $\mathcal{L}$.

(2) If μ is a variable z distinct from x, then we need to prove that

$$\Gamma \vdash_{\mathcal{L}} \exists_y (y_z = z \wedge s^*[y/y_s]) \leftrightarrow (y_z = z)$$

Since s^* is a functional formula of Γ with respect to y_s, $\Gamma \vdash_{\mathcal{L}} \exists_y s^*[y/y_s]$. Combining this with

$$\vdash_{\mathcal{L}} \exists_y (y_z = z \wedge s^*[y/y_s]) \leftrightarrow (y_z = z \wedge \exists_y s^*[y/y_s])$$

proves the assertion.

(3) Suppose that μ is of the form $g\mu_1...\mu_n$ where g is an n-ary function symbol in $\mathcal{L}$ and $\mu_1, ..., \mu_n$ are terms. The proof is essentially identical to the one of the next case.

(4) Suppose that μ is of the form $f\mu_1...\mu_l$ where $\mu_1, ..., \mu_l$ are terms. Then the formula $\mu^*[y_{\mu[s/x]}/y_\mu, y/x]$ is

$$\exists_{y_{\mu_1}}...\exists_{y_{\mu_l}} (Fy_{\mu_1}...y_{\mu_l}y_{\mu[s/x]} \wedge \mu_1^*[y/x] \wedge ... \wedge \mu_l^*[y/x])$$

We denote

$$\nu \equiv Fy_{\mu_1}...y_{\mu_l}y_{\mu[s/x]} \wedge \mu_1^*[y/x] \wedge ... \wedge \mu_l^*[y/x]$$

Then application of Proposition 4.7 implies

$$\vdash_{\mathcal{L}} \exists_{y_{\mu_1}}...\exists_{y_{\mu_l}} \nu \leftrightarrow \exists_{y_{\mu_1[s/x]}}...\exists_{y_{\mu_l[s/x]}} \nu[y_{\mu_1[s/x]}/y_{\mu_1}, ..., y_{\mu_l[s/x]}/y_{\mu_l}]$$

Since $y_{\mu_1[s/x]}, ..., y_{\mu_l[s/x]}$ are not free in $s^*[y/y_s]$, we derive that

$$\exists_{y_{\mu_1[s/x]}}...\exists_{y_{\mu_l[s/x]}} \nu[y_{\mu_1[s/x]}/y_{\mu_1}, ..., y_{\mu_l[s/x]}/y_{\mu_l}] \wedge s^*[y/y_s] \leftrightarrow$$

$$\leftrightarrow \exists_{y_{\mu_1[s/x]}}...\exists_{y_{\mu_l[s/x]}} \left(\nu[y_{\mu_1[s/x]}/y_{\mu_1}, ..., y_{\mu_l[s/x]}/y_{\mu_l}] \wedge s^*[y/y_s]\right)$$

is a theorem of minimal logic. Since

$$\nu[y_{\mu_1[s/x]}/y_{\mu_1}, ..., y_{\mu_l[s/x]}/y_{\mu_l}] \wedge s^*[y/y_s] \leftrightarrow$$

$$\leftrightarrow Fy_{\mu_1[s/x]}...y_{\mu_l[s/x]}y_{\mu[s/x]} \wedge \bigwedge_{i=1}^{k} \left(\mu_i^*[y_{\mu_i[s/x]}/y_{\mu_i}, y/x] \wedge s^*[y/y_s]\right)$$

is a theorem of minimal logic and s^* is functional formula in Γ with respect to y_s, we derive by Proposition 6.4 that

$$\exists_y \left(\nu[y_{\mu_1[s/x]}/y_{\mu_1}, ..., y_{\mu_l[s/x]}/y_{\mu_l}] \wedge s^*[y/y_s]\right) \leftrightarrow$$

$$\leftrightarrow Fy_{\mu_1[s/x]}...y_{\mu_l[s/x]}y_{\mu[s/x]} \wedge \bigwedge_{i=1}^{k} \exists_y \left(\mu_i^*[y_{\mu_i[s/x]}/y_{\mu_i}, y/x] \wedge s^*[y/y_s]\right)$$

is deducible from Γ in $\mathcal{L}$. By induction hypothesis

$$\Gamma \vdash_{\mathcal{L}} \exists_y \left(\mu_i^*[y_{\mu_i[s/x]}/y_{\mu_i}, y/x] \wedge s^*[y/y_s]\right) \leftrightarrow \mu_i[s/x]^*$$

for every $i = 1, ..., k$. Thus

$$\Gamma \vdash_{\mathcal{L}} \exists_y \left(\nu[y_{\mu_1[s/x]}/y_{\mu_1}, ..., y_{\mu_l[s/x]}/y_{\mu_l}] \wedge s^*[y/y_s]\right) \leftrightarrow$$

$$\leftrightarrow Fy_{\mu_1[s/x]}...y_{\mu_l[s/x]}y_{\mu[s/x]} \wedge \bigwedge_{i=1}^{k} \mu_i[s/x]^*$$

Note that

$$\mu[s/x]^* \equiv \exists_{y_{\mu_1[s/x]}}...\exists_{y_{\mu_l[s/x]}} \left(Fy_{\mu_1[s/x]}...y_{\mu_l[s/x]}y_{\mu[s/x]} \wedge \bigwedge_{i=1}^{k} \mu_i[s/x]^*\right)$$

Combining all these results together and using Proposition 4.8 we infer

$$\Gamma \vdash_{\mathcal{L}} \exists_y \left(\mu^*[y_{\mu[s/x]}/y_\mu, y/x] \wedge s^*[y/y_s]\right) \leftrightarrow \mu[s/x]^*$$

(5) Suppose that μ is an atomic formula of the form $r\mu_1...\mu_n$ where r is an n-ary relation symbol and $\mu_1,...\mu_n$ are terms, then the proof is analogous to the above.

(6) If μ has one of the forms

$$(\mu_1 \to \mu_2), (\mu_1 \wedge \mu_2), (\mu_1 \vee \mu_2)$$

for some formulas μ_1, μ_2, then the assertion for μ^* follows form Proposition 6.4.

(7) Suppose that μ is of the form $\forall_z \eta$ where z is a variable distinct from x and η is in $\mathfrak{U}$. Since by assumption s is substitutable in μ for x, we derive that z does not occur in s. Hence z does not occur in s^* and

$$\vdash_{\mathcal{L}} \exists_y (\forall_z \eta^*[y/x] \wedge s^*[y/y_s]) \leftrightarrow \exists_y \forall_z (\eta^*[y/x] \wedge s^*[y/y_s])$$

Proposition 6.5 and the fact that s^* is functional formula in Γ with respect to y_s imply that

$$\Gamma \vdash_{\mathcal{L}} \exists_y \forall_z (\eta^*[y/x] \wedge s^*[y/y_s]) \leftrightarrow \forall_z \exists_y (\eta^*[y/x] \wedge s^*[y/y_s])$$

Next by induction hypothesis we infer

$$\Gamma \vdash_{\mathcal{L}} \forall_z \exists_y (\eta^*[y/x] \wedge s^*[y/y_s]) \leftrightarrow \forall_z \eta[s/x]^*$$

Combining all these facts we deduce that

$$\Gamma \vdash_{\mathcal{L}} \exists_y (\forall_z \eta^*[y/x] \wedge s^*[y/y_s]) \leftrightarrow \forall_z \eta[s/x]^*$$

which is exactly

$$\Gamma \vdash_{\mathcal{L}} \exists_y (\mu^*[y/x] \wedge s^*[y/y_s]) \leftrightarrow \mu[s/x]^*$$

(8) If μ is of the form $\exists_z \eta$ where z is a variable distinct from x and η is in $\mathfrak{U}$, then the assertion follows from Proposition 4.8. The details are left for the reader.

(9) If μ is has one of the forms

$$\forall_x \eta, \exists_x \eta$$

for some η in $\mathfrak{U}$, then the assertion follows from the fact that s^* is functional formula in Γ.

Suppose that μ is a formula in $\mathfrak{U}$. Assume that s is a term in $\mathfrak{U}$ and x is a variable such that s is substitutable in μ for x and $\mu[s/x]$ is in $\mathfrak{U}$. We combine Fact 6 with

$$\vdash_{\mathcal{L}} \exists_y (\mu^*[y/x] \wedge s^*[y/y_s]) \to \exists_x \mu^*$$

and

$$\vdash_{\mathcal{L}} \forall_x \mu^* \to \exists_y (\mu^*[y/x] \wedge s^*[y/y_s])$$

for variable y not in V and distinct from y_t for all terms t in $\mathfrak{U}$. Thus we derive that

$$\Gamma \vdash_{\mathcal{L}} \mu[s/x]^* \to \exists_x \mu^*$$

and

$$\Gamma \vdash_{\mathcal{L}} \forall_x \mu^* \to \mu[x/s]^*$$

Now we show that $D_1^*, ..., D_m^*$ gives rise to a deduction of ϕ^* from Γ.

Suppose that D_k is any axiom of $\mathcal{L}[f]$. Since the translation $\mu \mapsto \mu^*$ preserves connectives and quantifiers, we deduce that D_k^* is an axiom of $\mathcal{L}$ provided that D_k is not an instance of universal instantiation or existential introduction. On the other hand the consequence of Fact 6 presented above implies that D_k^* is provable from Γ if it is an instance of universal instantiation or existential introduction.

Note also that

$$\left(\forall_{x_1}...\forall_{x_l} Fx_1...x_l f x_1...x_l\right)^* \equiv$$

$$\equiv \forall_{x_1}...\forall_{x_l} \exists_{y_{x_1}}...\exists_{y_{x_l}} \exists_{y_{fx_1...x_k}} \left(Fy_{x_1}...y_{x_l} y_{fx_1...x_l} \wedge y_{x_1} = x_1 \wedge ... \wedge y_{x_l} = x_l\right)$$

is provable from Γ in $\mathcal{L}$.

Finally, if $D_j \equiv (D_i \to D_k)$ for some $i, j < k$, then $D_j^* \equiv (D_i^* \to D_k^*)$.

It follows that

$$D_1^*, ..., D_m^*$$

is a deduction of ϕ^* from consequences of Γ in $\mathcal{L}$. Hence $\Gamma \vdash_{\mathcal{L}} \phi^*$. Since $\vdash_{\mathcal{L}} \phi^* \leftrightarrow \phi$ according to Fact 4, we deduce that ϕ is deducible from Γ in $\mathcal{L}$.

Since ϕ is arbitrary formula of $\mathcal{L}$ such that $\Gamma[f] \vdash_{\mathcal{L}[f]} \phi$, this completes the proof. □

Proof of the theorem. For each θ in $\mathcal{F}$ we introduce fresh relation symbol F_θ of the arity equal to the number of free variables of θ. The collection of these newly introduced relation symbols is denoted by $\mathcal{R}$. Let $\mathcal{M}$ be a first-order language obtained from $\mathcal{L}[\mathcal{F}]$ by adding F_θ for each θ in $\mathcal{F}$. Next we define Δ to be the collection of formulas of $\mathcal{M}$ obtained from Γ by adding

$$\forall_{x_1}...\forall_{x_l}\forall_{y_\theta}\left(\theta \leftrightarrow F_\theta x_1...x_l y_\theta\right),\ \forall_{x_1}...\forall_{x_l} F_\theta x_1...x_l f_\theta x_1...x_l$$

for each θ in $\mathcal{F}$. Above we assume that $x_1, ..., x_l, y_\theta$ are all variables that occur free in θ and that

$$\Gamma \vdash_{\mathcal{L}} \forall_{x_1}...\forall_{x_l}\exists!_{y_\theta}\,\theta$$

for every θ in $\mathcal{F}$.

Suppose that ϕ is a formula of $\mathcal{L}$ such that $\Gamma[\mathcal{F}] \vdash_{\mathcal{L}[\mathcal{F}]} \phi$. First note that if ψ is a formula in $\Gamma[\mathcal{F}]$, then $\Delta \vdash_{\mathcal{M}} \psi$. Hence $\Delta \vdash_{\mathcal{M}} \phi$. There are only finitely many newly introduced relation and function symbols which are involved in some fixed deduction of ϕ from Δ in $\mathcal{M}$. Suppose that these are

$$F_{\theta_1}, ..., F_{\theta_n}, f_{\theta_1}, ..., f_{\theta_n}$$

for some $\theta_1, ..., \theta_n$ in $\mathcal{F}$. First by repeated application of Lemma 7.1.1 we may eliminate $f_{\theta_1}, ..., f_{\theta_n}$. Next by Theorem 5.4 we derive that $\Gamma \vdash_{\mathcal{L}} \phi$. Since ϕ is arbitrary formula of $\mathcal{L}$ such that $\Gamma[\mathcal{F}] \vdash_{\mathcal{L}[\mathcal{F}]} \phi$, the proof is completed. □

8. Intuitionistic and classical logic

In this section we fix a first-order language $\mathcal{L}$.

Definition 8.1. If we add the following propositional axiom schema to the propositional axioms of minimal logic $\bot \to \phi$ for every formula ϕ of $\mathcal{L}$, we obtain the propositional axiom schemas of *intuitionistic logic*.

Definition 8.2. If we add the following propositional axiom schema to the propositional axioms of minimal logic $(\neg(\neg\phi)) \to \phi$ for every formula ϕ of $\mathcal{L}$, we obtain the propositional axiom schemas of *classical logic*.

Fact 8.3. *Let Γ be a collection of formulas of $\mathcal{L}$ and let ϕ be a formula of $\mathcal{L}$. If ϕ is deducible from Γ in minimal system, then it is also deducible from Γ in intuitionistic and classical systems.*

Proof. Follows immediately from definitions of minimal, intuitionistic and classical systems. □

Now we prove famous *ex falso sequitur quodlibet*. It was first obtained by medieval logicians in classical scholastic period.

Fact 8.4. *Let ϕ be a formula of $\mathcal{L}$. Then all generalizations of $\bot \to \phi$ are provable in classical logic.*

Proof. Consider the following sequence of formulas of $\mathcal{L}$.

1. $\bot$
2. $\bot \to ((\phi \to \bot) \to \bot)$

3. $(\phi \to \bot) \to \bot$

4. $((\phi \to \bot) \to \bot) \to \phi$

5. ϕ

Note that **1** is an axiom. Next **3** is obtained by modus ponens from **1** and **2**. Observe that **4** is axiom of classical logic. Then **5** is obtained by modus ponens from **3, 4**. Thus the sequence above is a deduction of ϕ from $\bot$ from in classical calculus. Applying Theorem 4.5 we infer $\bot \to \phi$ is deducible in classical logic. Next by Theorem 4.6 we derive that all generalizations of $\bot \to \phi$ are provable in classical logic. This completes the proof. □

Corollary 8.5. *Let* Γ *be a collection of formulas of* $\mathcal{L}$ *and let* ϕ *be a formula of* $\mathcal{L}$*. If* ϕ *is deducible from* Γ *in inutitionistic system, then it is also deducible from* Γ *in classical system.*

Proof. It is immediate consequence of Fact 8.4. □